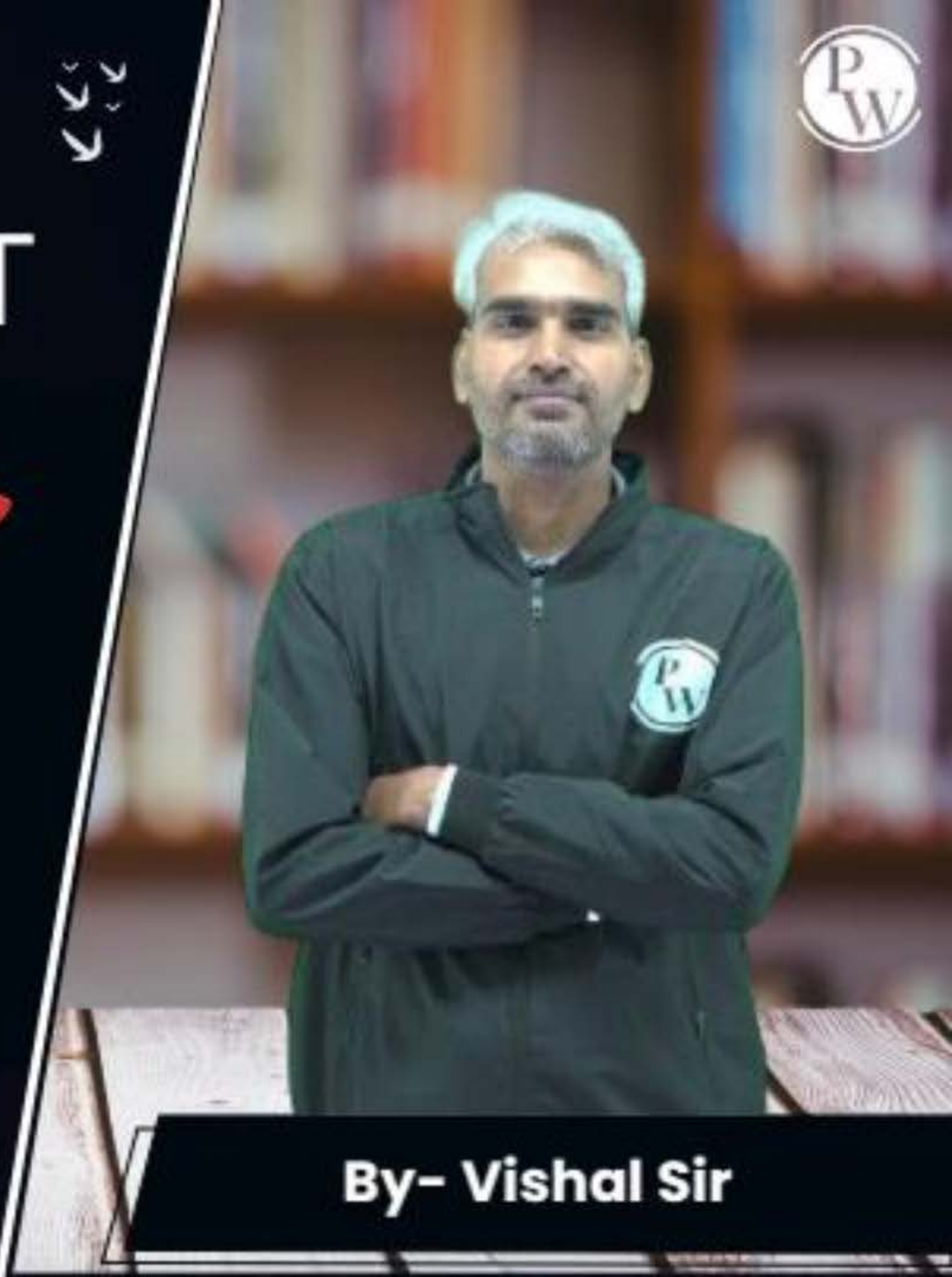
Computer Science & IT



Discrete Mathematics

Graph Theory

Lecture No. 01



By- Vishal Sir

Recap of Previous Lecture



Topic Cyclic group

Topic Some important properties

Slide

Topics to be Covered

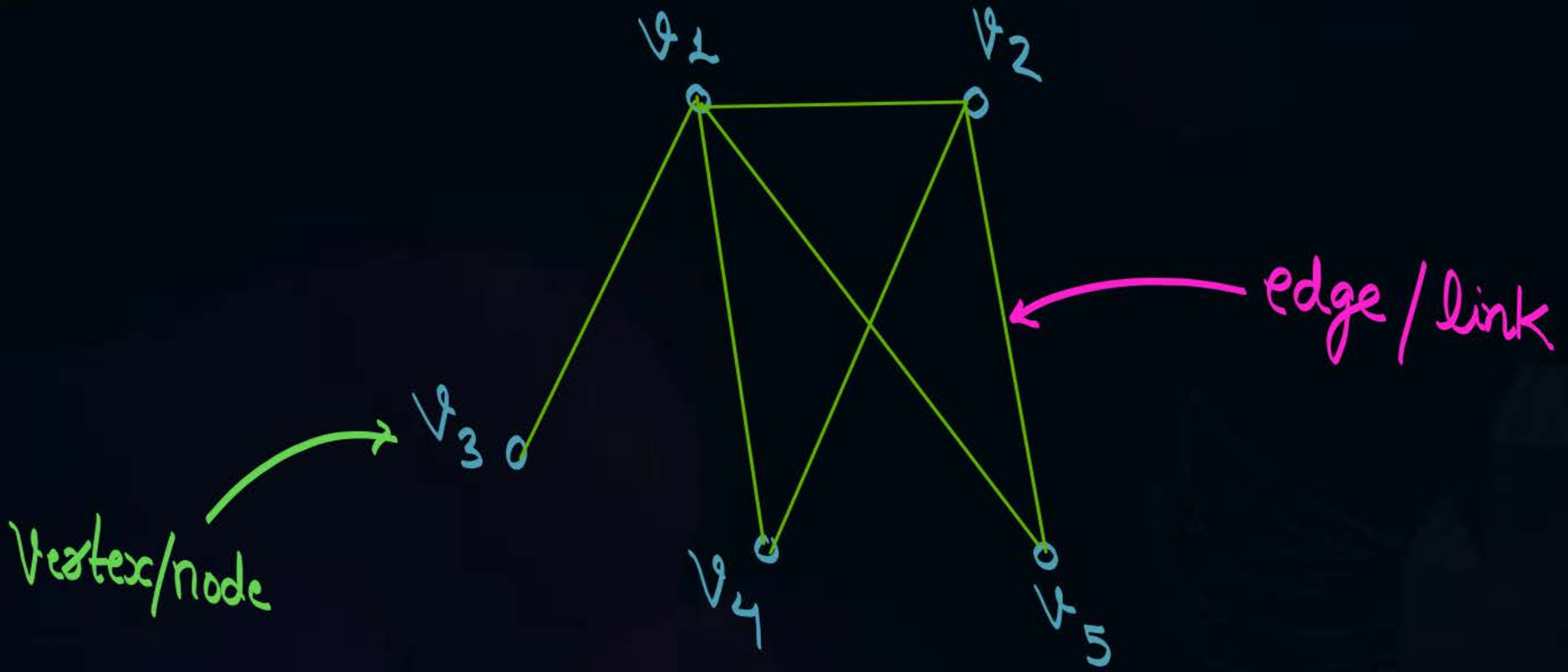


- Topic: Representation of graph
- Topic: Simple graph, multi graph and pseudo graph
- Topic: Maximum number of edges in a simple graph
- Topic: Maximum number of simple graph with n vertices
- Topic: Terminologies related to graphs

Slide



Topic : Graph





Topic : Graph

order will matter

A Graph is defined as an ordered pair of two sets,

i.e., $G = (V, E)$

Name assigned to the graph

Set of Vertices

Set of edges

$|V|$ = No. of vertices (Order of graph G)

$|E|$ = No. of edges (Size of graph G)



Topic : Representation of edge

In a non-directed graph an edge is represented by a set of two vertices

e.g., $\{v_i, v_j\}$ which represents an edge between vertices v_i and v_j

$\{v_j, v_i\}$

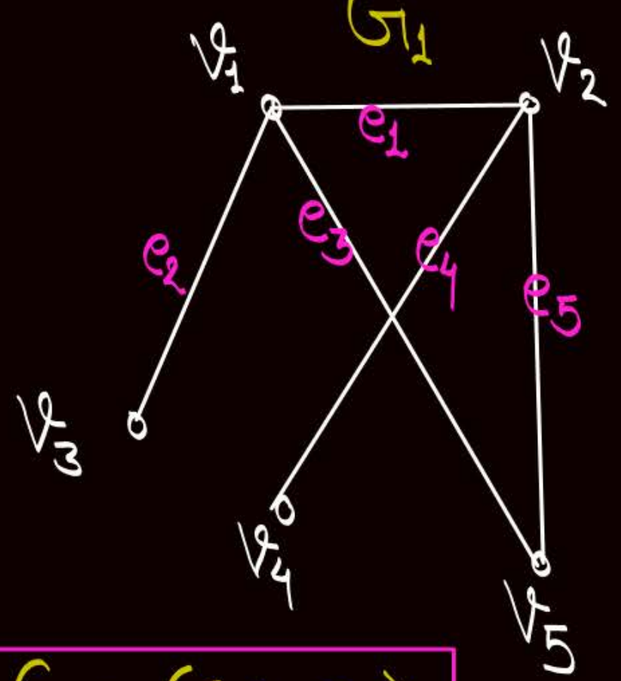
v_i & v_j are end vertices associated with the edge.

In a directed graph an edge is represented by an order pair of two vertices

e.g., (v_i, v_j) which represents an edge from vertex v_i to vertex v_j

(v_j, v_i)

4 Undirected graph



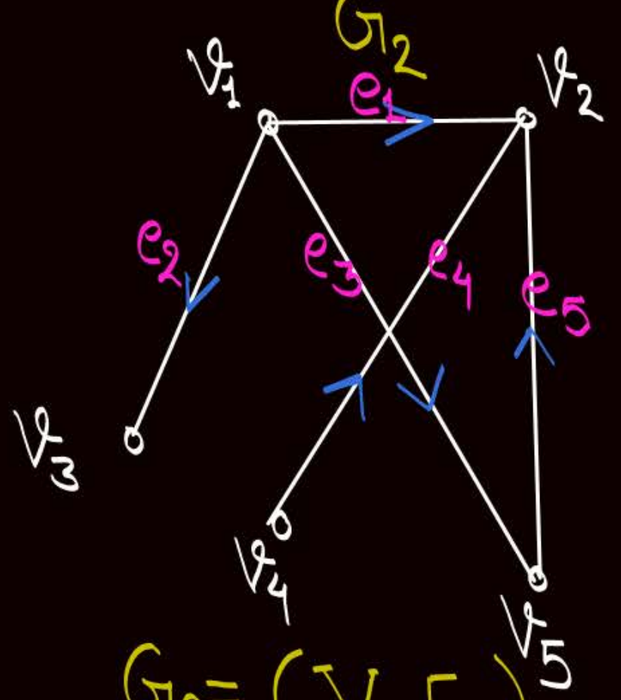
$G_1 = (V, E)$

$V = \{v_1, v_2, v_3, v_4, v_5\}$

$E = \{e_1, e_2, e_3, e_4, e_5\}$

$E = \{\{v_1, v_2\}, \{v_1, v_3\}, \{v_1, v_5\}, \{v_2, v_4\}, \{v_2, v_5\}\}$

5 Directed graph



$G_2 = (V, E)$

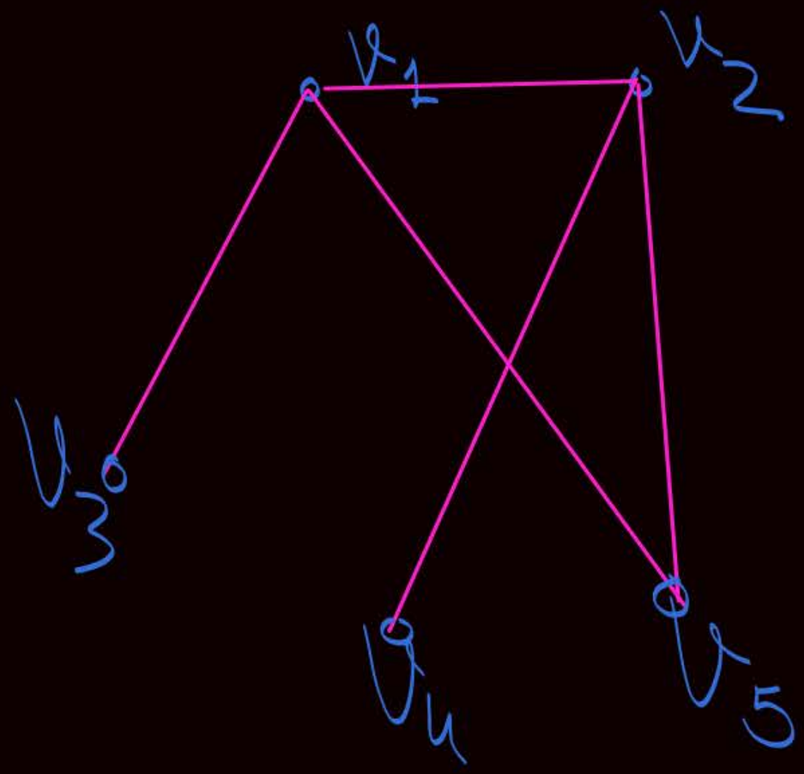
$V = \{v_1, v_2, v_3, v_4, v_5\}$

$E = \{e_1, e_2, e_3, e_4, e_5\}$

$E = \{(v_1, v_2), (v_1, v_3), (v_1, v_5), (v_4, v_2), (v_5, v_2)\}$

$$G_1 = (\{v_1, v_2, v_3, v_4, v_5\}, \{e_1, e_2, e_3, e_4, e_5\})$$

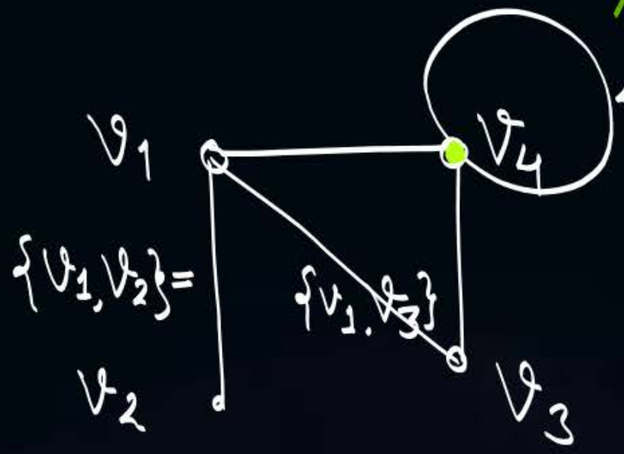
$$G_1 = (\{v_1, v_2, v_3, v_4, v_5\}, \{\{v_1, v_2\}, \{v_1, v_3\}, \{v_1, v_5\}, \{v_3, v_4\}, \{v_2, v_5\}\})$$



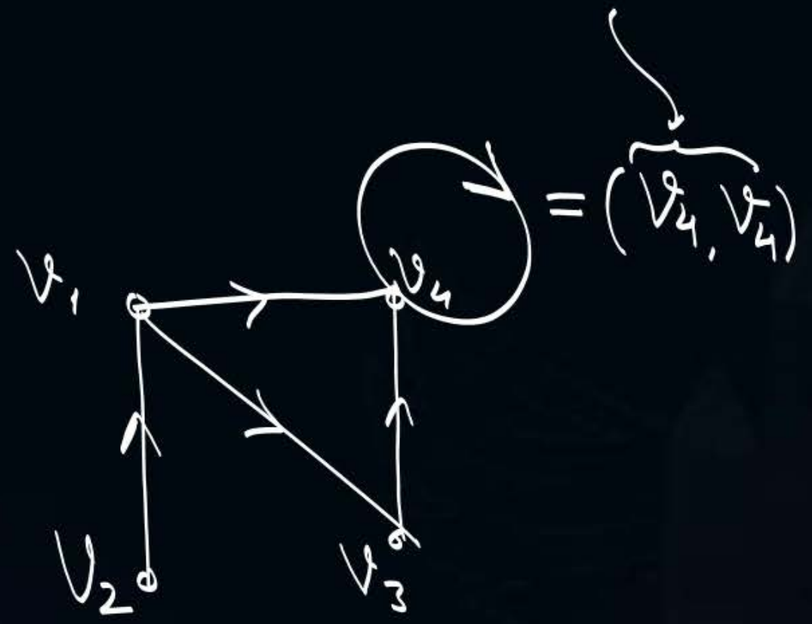


Topic : Loop

Self loop



$\{v_4, v_4\}$: Loop is an edge in which both "end vertices" are same



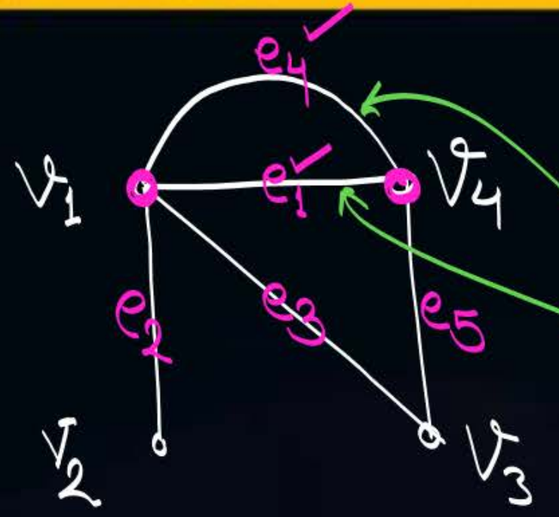


Topic : Parallel Edges

If two or more edge are associated with same end vertices, then edges are called parallel edges and resulting graph is called multi-graph.



Topic : Parallel Edges



$e_1 = \{v_1, v_4\}$

$e_4 = \{v_1, v_4\}$

} Both e_1 & e_4 are associated with same end vertices

$\therefore$

Parallel edges



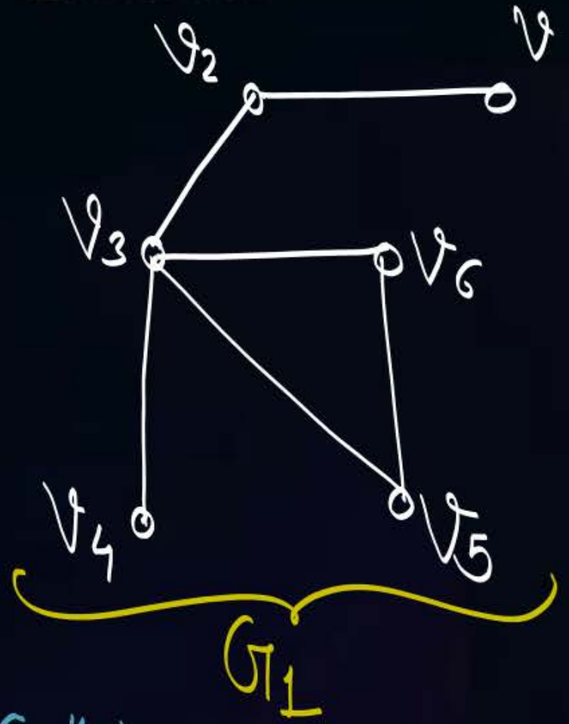
Topic : Simple graph

A graph with no self-loop and no parallel edge is called simple graph.

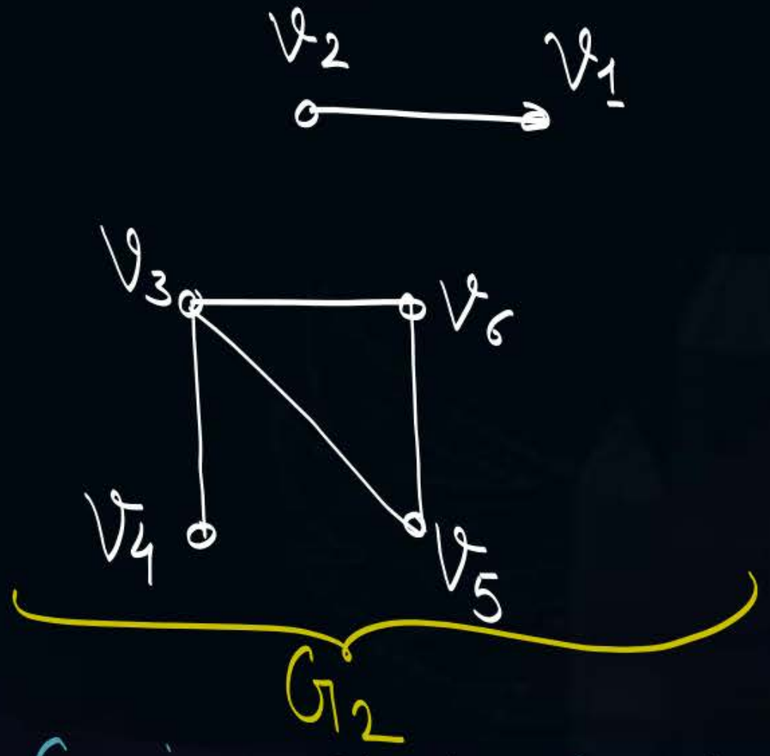


Topic : Simple graph

A graph with no self-loop and no parallel edge is called simple graph.



"G_1" is a simple graph as well as a connected graph



G_2 is a simple graph but G_2 is a disconnected graph



Topic : Classification of graph

	Loop	Parallel edge
Simple Graph	Not allowed ✗	Not allowed ✗
Multi-graph	Not allowed ✗	Allowed ✓
Pseudo graph	Allowed ✓	Allowed ✓



Topic : Simple Graph

Maximum number of edges possible in “Simple Graph” With n-vertices



Topic : Simple Graph

Maximum number of edges possible in "Simple Graph" With n-vertices

In a simple graph two distinct vertices are required to define an edge.

Maximum number of edges possible in a simple graph with n-vertices = No. of ways in which two distinct vertices can be chosen out of n-vertices

$$= {}^{n}C_2 = \frac{n(n-1)}{2} = \frac{n^2-n}{2}$$



Topic : Simple Graph

Total Number of "Simple Graphs" possible With 'n' labeled vertices







Topic : Simple Graph

Total **Number of "Simple Graphs" possible With 'n' labeled vertices**

Each simple graph with 'n' vertices will contain all 'n' vertices and at most $\left(\frac{n^2-n}{2}\right)$ edges {all simple}

i.e. 0 to $\left(\frac{n^2-n}{2}\right)$ edges



Topic : Simple Graph

Total Number of "Simple Graphs" possible With 'n' labeled vertices

Each simple graph with 'n' vertices will contain all 'n' vertices and at most $\left(\frac{n^2-n}{2}\right)$ edges {all simple}

i.e.,

No. of simple graphs Possible with n-vertices = (Choose all 'n' out of 'n' vertices) and (Choose 0 or more edges out of $\left(\frac{n^2-n}{2}\right)$ simple edges)

$$= {}^{n}C_{n} * \left({}^{\frac{n^2-n}{2}}C_0 + {}^{\frac{n^2-n}{2}}C_1 + {}^{\frac{n^2-n}{2}}C_2 + \cdots + {}^{\frac{n^2-n}{2}}C_{\frac{n^2-n}{2}}\right)$$

$$= 1 * \text{——— / ———}$$

$$\therefore \quad {}^{n}C_0 + {}^{n}C_1 + {}^{n}C_2 + \cdots + {}^{n}C_n = 2^n$$

$$\therefore \quad {}^{\frac{n^2-n}{2}}C_0 + {}^{\frac{n^2-n}{2}}C_1 + {}^{\frac{n^2-n}{2}}C_2 + \cdots + {}^{\frac{n^2-n}{2}}C_{\frac{n^2-n}{2}} = 2^{\left(\frac{n^2-n}{2}\right)} = 2^{({}^{n}C_2)}$$



Topic : Simple Graph

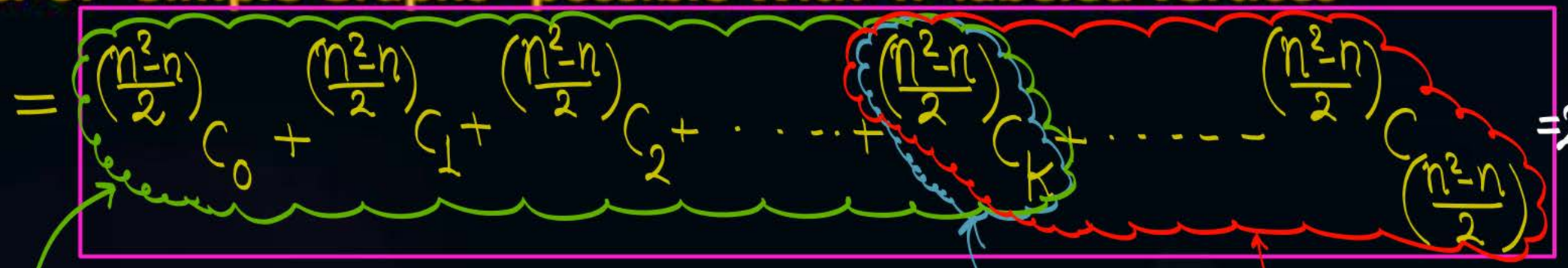
Total **Number of "Simple Graphs" possible With 'n' labeled vertices**

$$= \left(\binom{\frac{n^2-n}{2}}{C_0} + \binom{\frac{n^2-n}{2}}{C_1} + \binom{\frac{n^2-n}{2}}{C_2} + \cdots + \binom{\frac{n^2-n}{2}}{C_{\left(\frac{n^2-n}{2}\right)}} \right)$$

$$= \boxed{2^{\left(\frac{n^2-n}{2}\right)} = 2^{{}^{n}C_2} = 2^{\frac{n(n-1)}{2}}}$$

Topic : Simple Graph

Number of "Simple Graphs" possible With 'n' labeled vertices

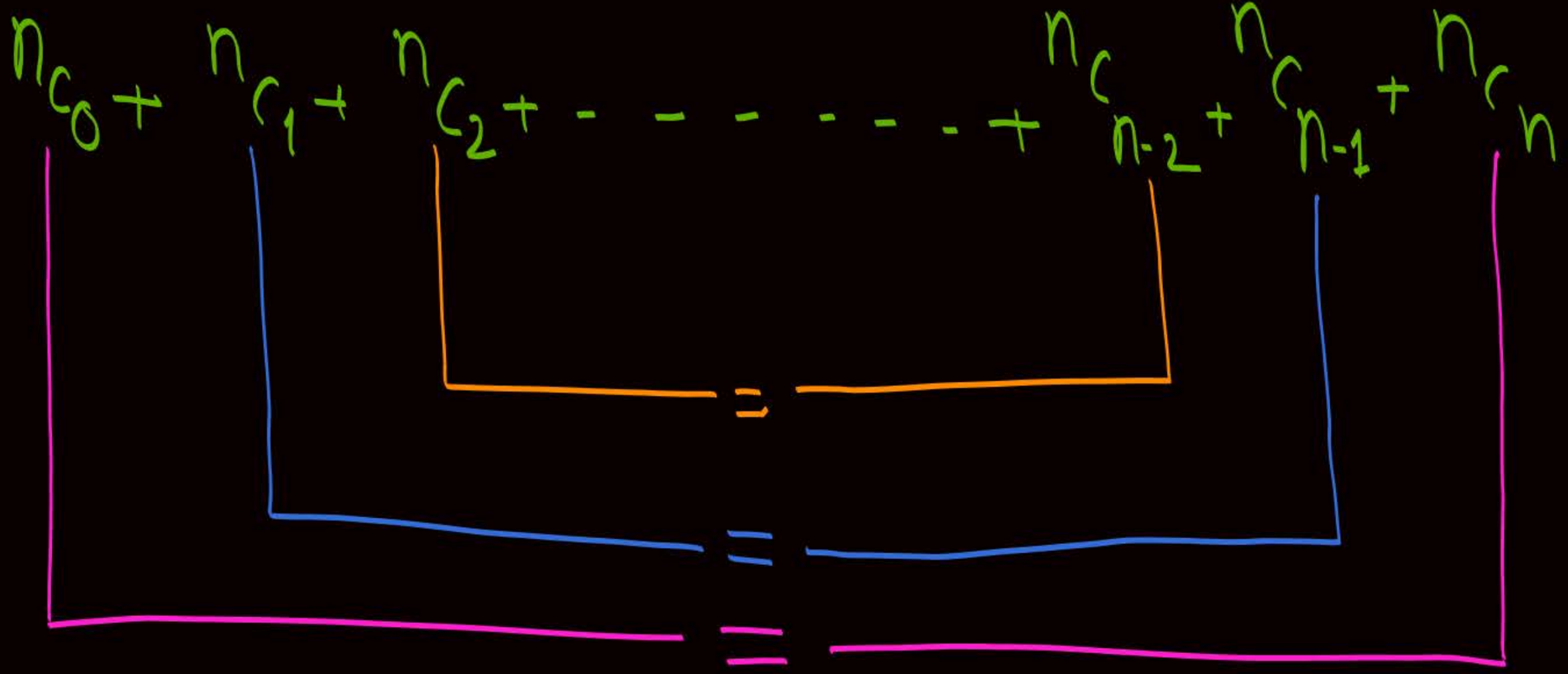


No. of simple graphs possible with n-vertices and at most 'K' edges

No. of simple graphs possible with n-vertices and Exactly 'K'- edges

No. of simple graphs Possible with 'n' vertices and at least 'K' edges

We know $nC_r = nC_{n-r}$



In the Sequence,

Summation of first 'k' terms = Summation of last 'k' terms

$$\therefore \quad {}^{n}C_0 + {}^{n}C_1 + {}^{n}C_2 + \cdots + {}^{n}C_n = 2^n$$

$$\therefore \quad {}^{\frac{n^2-n}{2}}C_0 + {}^{\frac{n^2-n}{2}}C_1 + {}^{\frac{n^2-n}{2}}C_2 + \cdots + {}^{\frac{n^2-n}{2}}C_{\frac{n^2-n}{2}} = 2^{\left(\frac{n^2-n}{2}\right)} = 2^{({}^{n}C_2)}$$

No. of simple graphs possible with at most 'K' edges = No. of simple graphs with at least $\left(\frac{n^2-n}{2} - K\right)$ edges

i.e. Summation of 1st "K+1" terms = Summation of last (K+1) terms

Q:- How many simple graphs are possible with '5' labeled vertices.

$$= 2^{5C_2} = 2^{10} = \underline{1024}$$

Q: How many simple graphs are possible with '4' labeled vertices and at most '2' edges ?

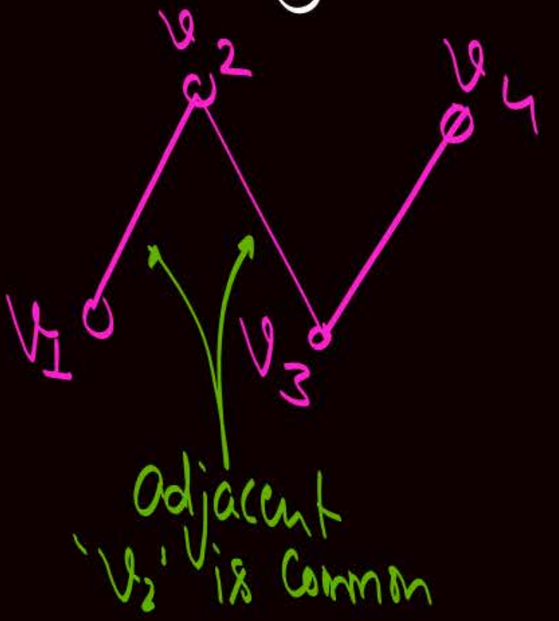
↳ With '4' labeled vertices maximum "$^4C_2 = 6$" edges are possible in a simple graph

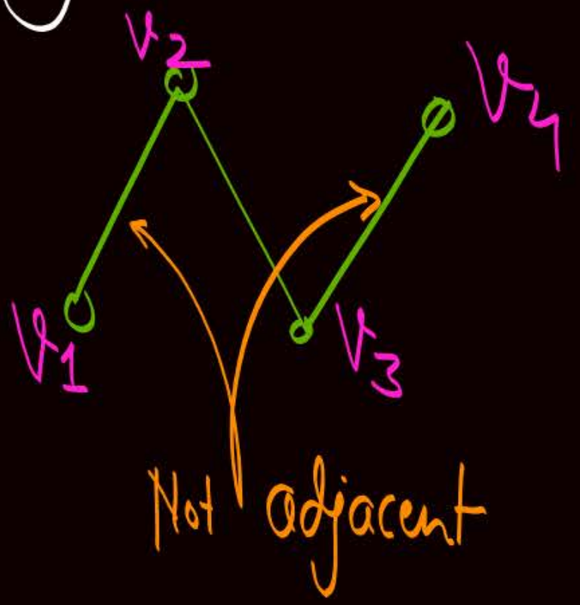
∴ Total no. of simple graphs possible with '4' vertices & at most '2' edges $= \left({}^6C_0 + {}^6C_1 + {}^6C_2 \right)$

$= 1 + 6 + 15 = \boxed{22}$ Ans

- **Adjacent vertices** :- Two vertices are said to be adjacent if there exist a direct edge between them

- **Adjacent edges** :- Two edges are said to be adjacent if they share any common vertex.







2 mins Summary

Topic Representation of graph

Topic Simple graph, multi graph and pseudo graph

Topic Maximum number of edges in a simple graph

Topic Maximum number of simple graph with n vertices

Topic Terminologies related to graphs

Slide

THANK - YOU